The Basics of Clustering

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- **Clustering**

The process of examining a collection of “points”,

and grouping the points into “clusters” according to some **distance measures**, which are the same as “similiarity”.

- Points in the same cluster => A small distance from one another.
- Points in the different cluster => A large distance from one another.

Since we cluster data with no background knowledges, this can be regarded as “Unsupervised Learning”.

- **A Goal of this Chapter : “How to cluster data?”**

- Offer methods for discovering clusters in data
- Learn how to manage and cluster the very big data

- **Points**

A dataset suitable for clustering is a collection of “points”, which are objects belong to some “space”.

- **Space**

A space is just a <u>universal set of points</u>.

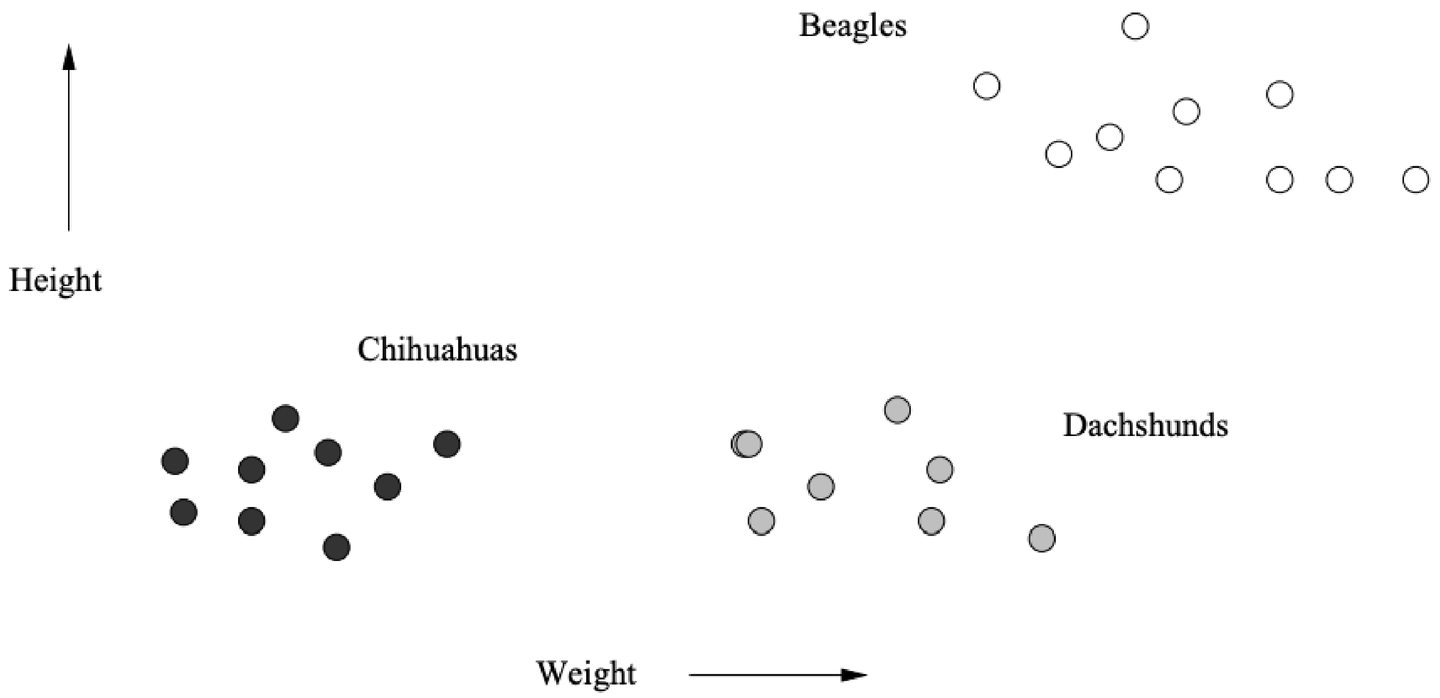


Figure 7.1: Heights and weights of dogs taken from three varieties

1. Introduction to Clustering Techniques

- **Case of a Euclidean Space**

In particular, a Euclidean space's points are vectors of real numbers.

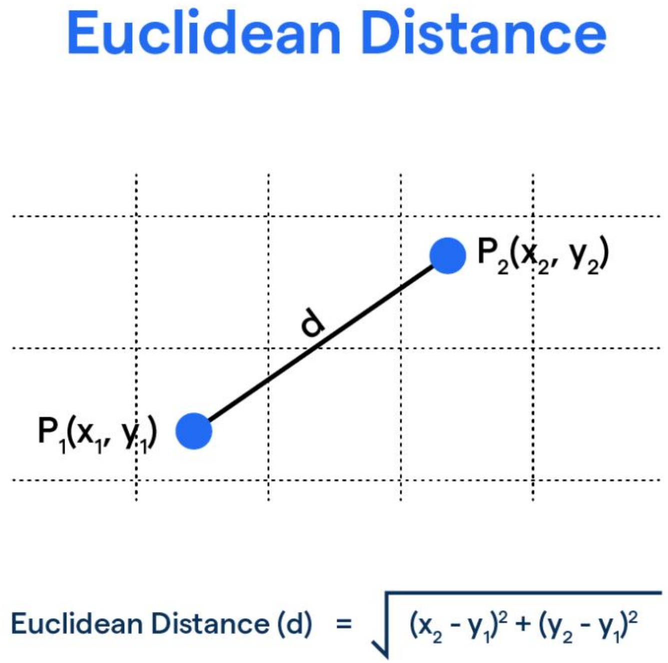
- The length of the vector = The number of dimensions of the space

- Components of the vector = "coordinates" of the represented points

- **Euclidean Distance d(A,B)**

for A(x1, y1) and B(x2, y2),

$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



1. Introduction to Clustering Techniques

- **Manhattan Distance d(A, B)**

: Sum of the magnitudes of the differences in each dimension

for A(x1, y1) and B(x2, y2),

d(A, B) = $|x2 - x1| + |y2 - y1|$

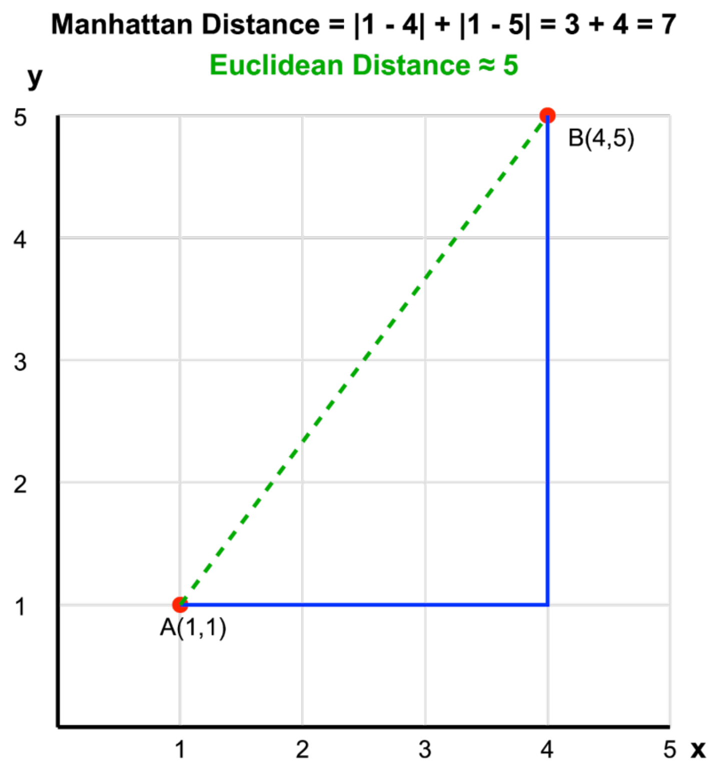
- Example of distance measures for non-Euclidean space

① Jaccard Distance

② Cosine Distance

③ Hamming Distance

④ Edit Distance



- **Important Requirements in Distance Measures**

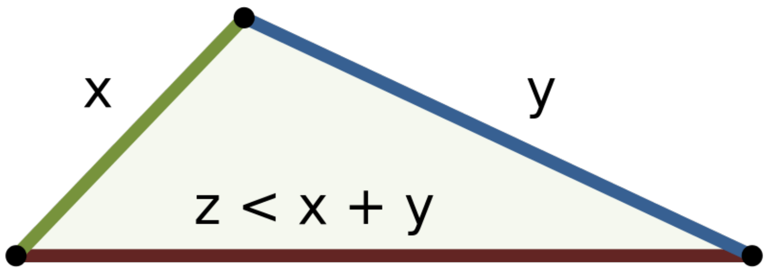
① Distances are always non - negative (>= 0).

② Distance ‘0’ is the only the distance between a point and itself.

③ Distance is symmetric.

Measuring order doesn’t matter.

④ Distance measures obey the triangle inequality.



- **Clustering Strategies**

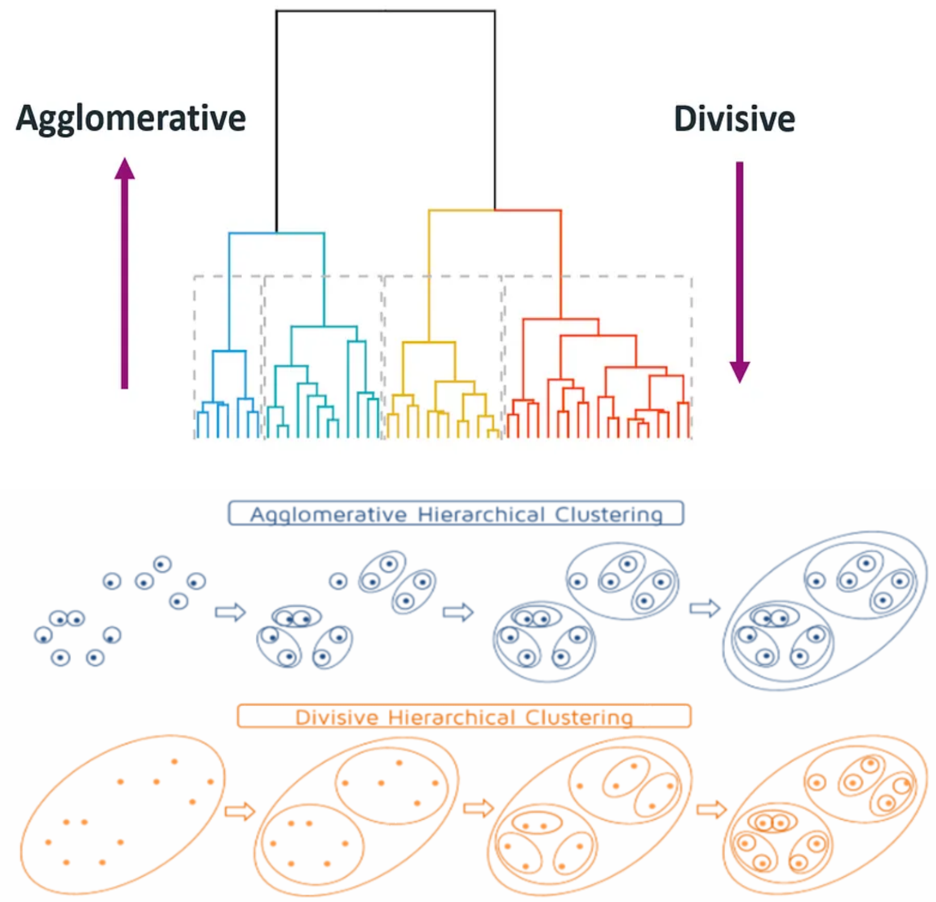
Two fundamentally different strategies in clustering algorithms

① Hierarchical / agglomerative algorithms

=> Clusters are combined based on “closeness”

② Point Assignment

=> Points are assigned to the cluster into which it best fits.



1. Introduction to Clustering Techniques

- **Clustering Strategies**

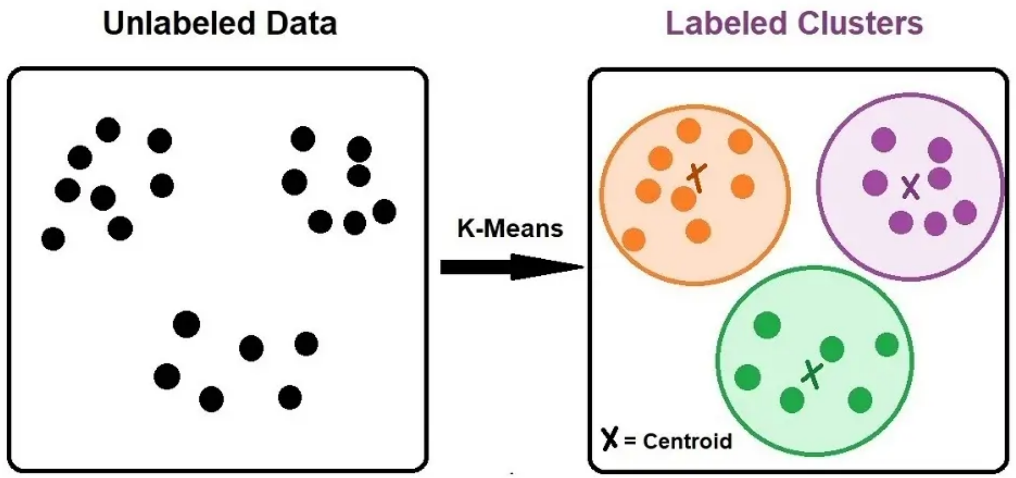
Another way to distinguish algorithms for clustering

① In **Euclidean Space**, it is possible to summarize a collection of points by their **centroids** (= the average of the points).

② In **non - Euclidean Space**, there is no notion of a centroid.

=> Have to develop another way to summarize clusters

③ Algorithms for large amounts of data often must take shortcuts, since it is infeasible to look at all pairs of points.

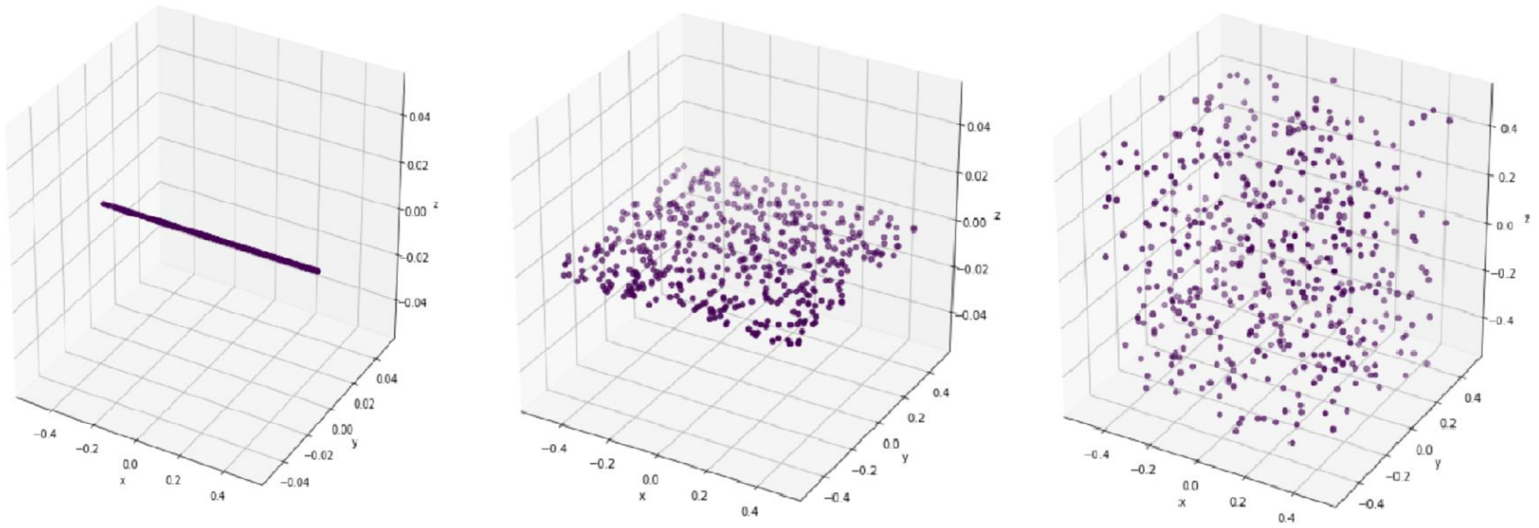


2. The Curse of Dimensionality

- In high dimensions, almost all pairs of points **<u>are equally far away</u>** from one another.

- Almost any two vectors are almost orthogonal.

=> As the number of dimension increases, **<u>the harder it gets to generate meaningful models</u>**.



2. The Curse of Dimensionality

▪ The Distribution of Distances in a High - Dimensional Space

In a d - dimensional Euclidean space, the distance between to points $\mathbf{x} = [x_1, x_2, \ldots, x_d]$ and $\mathbf{y} = [y_1, y_2, \ldots, y_d]$

=> $\text{Distance}(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{d} (x_i - y_i)^2}$

- Each xi and yi is a random variable chosen uniformly in the range [0, 1].
- If d = 1, we are placing random points on a line of length 1.
- Since d is large, we can expect that some $|x_i - y_i| \approx 1$

  => It becomes the **lower bound** of 1 on the distance between almost any two random points.

- The maximum possible distance occurs when $|x_i - y_i| = 1$ for all i,

=> $\text{Max Distance} = \sqrt{d}.$

- **Angles Between Vectors**

In large d, let's assume that we have 3 points A, B and C in a d - dimensional space.

(Do not assume points are in the unit cube; they can be anywhere in the space.)

$\mathbf{A} = [x_1, x_2, \ldots, x_d]$, $\mathbf{B} = \mathbf{0}$ (the origin), and $\mathbf{C} = [y_1, y_2, \ldots, y_d]$

=> $\cos(\angle ABC) = \dfrac{\sum_{i=1}^{d} x_i y_i}{\sqrt{\sum_{i=1}^{d} x_i^2} \cdot \sqrt{\sum_{i=1}^{d} y_i^2}}$

① Denominator (Product of Magnitudes)

$$\sqrt{\sum_{i=1}^{d} x_i^2} \cdot \sqrt{\sum_{i=1}^{d} y_i^2}$$

- Each term x_i^2 and y_i^2 is positive, and their sum grows linearly with d.
- As $d \to \infty$, the magnitudes of the vectors $\mathbf{A}$ and $\mathbf{C}$ grow approximately as $\sqrt{d}$.

2. The Curse of Dimensionality

$\mathbf{A} = [x_1, x_2, \ldots, x_d]$, $\mathbf{B} = \mathbf{0}$ (the origin), and $\mathbf{C} = [y_1, y_2, \ldots, y_d]$

$$\cos(\angle ABC) = \frac{\sum_{i=1}^{d} x_i y_i}{\sqrt{\sum_{i=1}^{d} x_i^2} \cdot \sqrt{\sum_{i=1}^{d} y_i^2}}$$

② Numerator (Dot Product) $\displaystyle\sum_{i=1}^{d} x_i y_i$

- Each term $x_i y_i$ is the product of two independent random variables.
- Assuming x_i and y_i are chosen randomly from a symmetric distribution (e.g., uniform distribution centered around 0 or standard normal distribution):
  - The expected value of $x_i y_i$ is 0.
  - The variance of $x_i y_i$ is proportional to the variance of x_i and y_i, which are constant.
  - For d terms, the total sum of products will have an expected value of 0 and a standard deviation that scales as $\sqrt{d}$.

=> $\cos(\angle ABC) \approx \frac{\text{numerator}}{\text{denominator}} \approx 0$ as $d \to \infty$

(The numerator scales as $\sqrt{d}$, while the denominator scales as d.)

=> Since $\cos(\angle ABC) \approx 0$, the angle $\angle ABC$ is approximately $90°$.

Random vectors in high-dimensional spaces are nearly orthogonal.

- **Hierarchical Clustering in a <u>Euclidean Space</u>**

- Larger clusters will be constructed by combining two smaller clusters.

Considerations

① How will clusters be represented?

② How will we choose which two clusters to merge?

③ When will we stop combining cluster?

Clustering in a Euclidean Space

① We can set a centroid in a Euclidean Space.

(Which means we can initialize the clusters.)

② We can then use the merging rule.

(The distance between any two clusters == The distance between their centroids)

3. Hierarchical Clustering

- Example : 2 - dimensional Euclidean Space

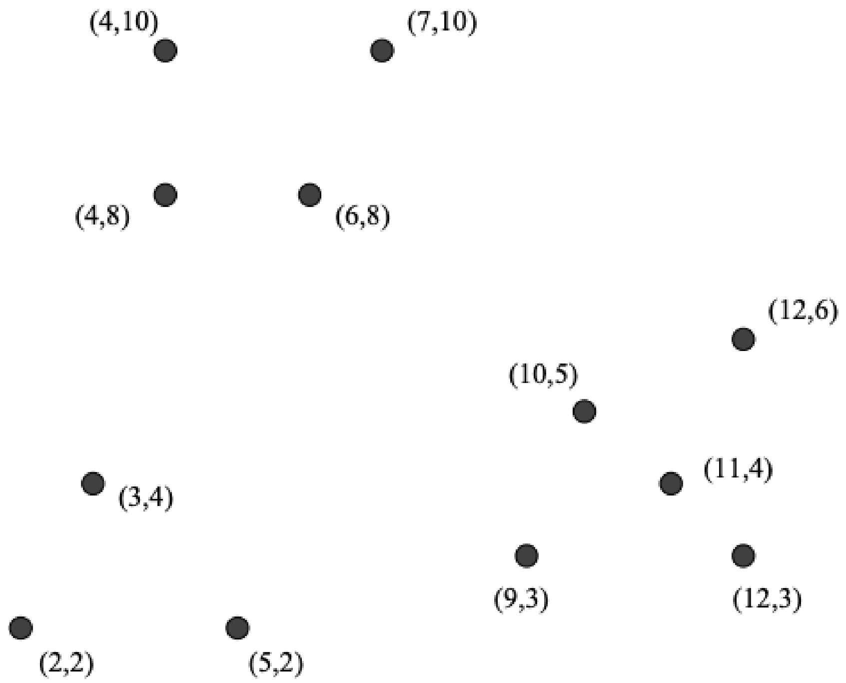


Figure 7.2: Twelve points to be clustered hierarchically

1st, each point is in a cluster by itself

=> each point is the centroid of that cluster

2nd, for two closest pairs (10, 5) - (11,4) / (11, 4) - (12, 3)

=> we can make a cluster with [(11, 4) - (12, 3)]

where centorid is (11.5, 3.5)

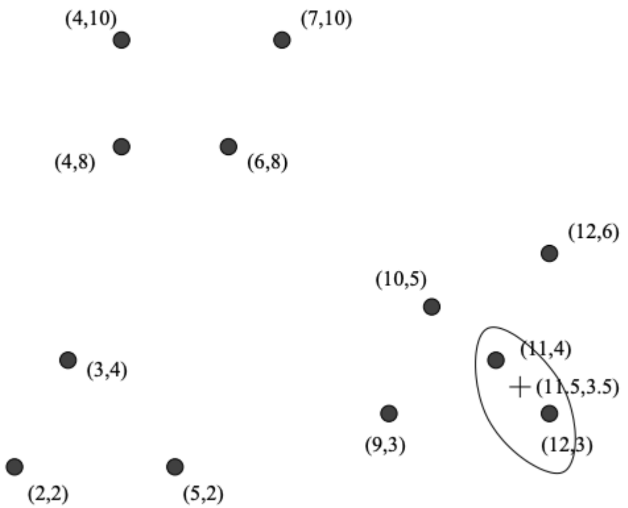


Figure 7.3: Combining the first two points into a cluster

3. Hierarchical Clustering

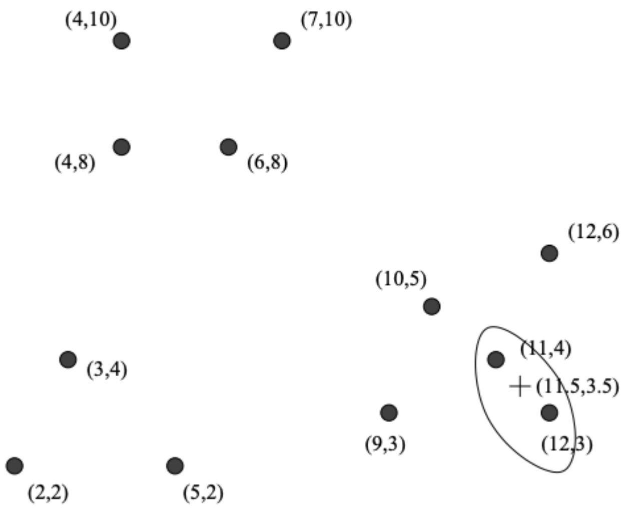


Figure 7.3: Combining the first two points into a cluster

=>

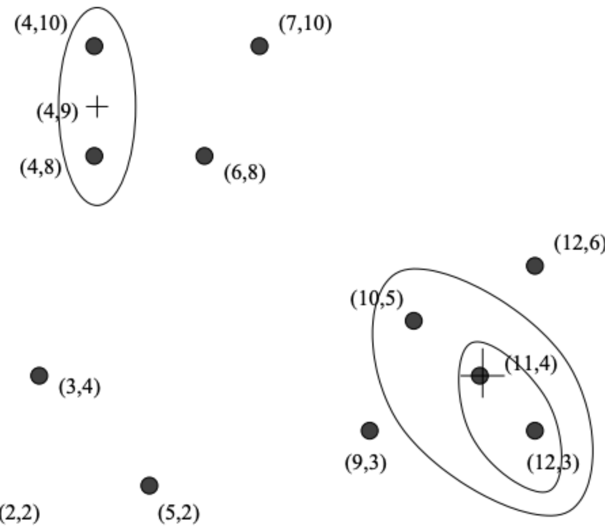


Figure 7.4: Clustering after two additional steps

=>

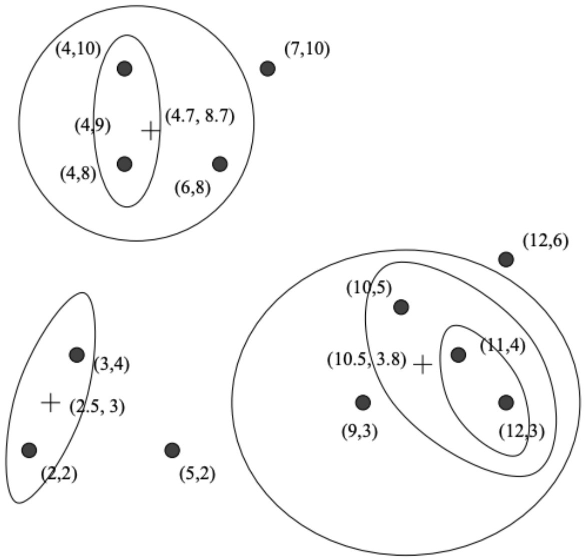


Figure 7.5: Three more steps of the hierarchical clustering

3rd, After creating a cluster with [(11, 4) - (12, 3)]

we can combine (10,5) with the new cluster next => [(10, 5) - (11, 4) - (12, 3)] with centroid (11, 4)

4th, we can combine [(4, 8) - (4, 10)] into one cluster with centroid (4, 9)

5th, (6, 8) is combined with [(4, 8) - (4, 10)]

6th, (2, 2) is combined with (3, 4)

7th, (9, 3) is combined with [(10, 5) - (11, 4) - (12, 3)]

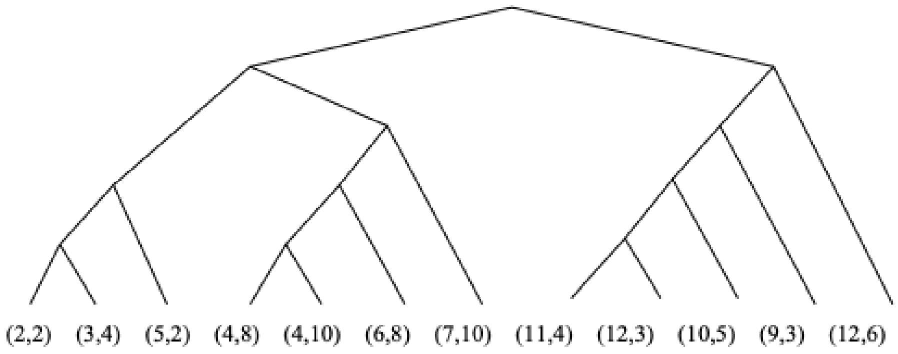


Figure 7.6: Tree showing the complete grouping of the points of Fig. 7.2

- **The Stopping Criteria for Clustering Process**

When shall we stop clustering?

① Stopping with a given number of clusters

- We decide the number of clusters beforehand, based on domain knowledge or external information.

② Stopping based on Cluster Quality

- The clustering process halts when further combinations or splits produce inadequate clusters.

③ Stopping at a Single Cluster with hierarchical tree

- The clustering process continues until all points are combined into a single cluster.

- The output is a dendogram (or a hierarchical tree).

4. Efficiency of Hierarchical Clustering

- **The Basic Algorithm for Hierarchical Clustering is not very Efficient.**

- At each step, we must compute the distances between each pair of clusters

- The initial step's time : $O(n^2)$ <= $\frac{n(n-1)}{2}$ unique pairs of points)

- Subsequent steps take time proportional to $(n-1)^2, (n-2)^2, \ldots$.

=> The sum of squres up to n : $O(n^3)$

- **More Efficient Implementation**

① The initial step's time is same as $O(n^2)$

② Priority Queue for Pair Distances $O(n^2)$

- Store the computed distances in a priority queue (min - heap)

- Each insertion into the priority queue takes $O(\log n)$

4. Efficiency of Hierarchical Clustering

③ Merging Clusters

When two clusters C and D are merged, we need to :

- Remove all the entries involving either C or D from the priority queue.

- Complexity : $O(n \log n)$

  - Since there are most 2n deletions (for each of the n clusters)

    and deleting an entry from the priority queue takes $O(\log n)$.

=> Total Complexity

For ①, their combined complexity is $O(n^2)$.

For ② and ③, these steps are repeated at most n times each with complexity of $O(n \log n)$.

$$O(n^2) + n \times O(n \log n) = O(n^2 \log n)$$

(It is better than $O(n^3)$, but still becomes infeasible for very large n.)

5. Alternative Rules for Controlling Hierarchcial Clustering

- **Another <u>Option 1</u> for Hierarchical Clustering**

① Take the distances between two clusters to be the minimum of the distances between any two points, one chosen from each cluster.

② Take the distance between two clusters to be the average distance of all pairs of points, one from each cluster.

③ The **<u>radius</u>** of a cluster is the maximum distance between all the points and the centroids.
Combine two clusters whose resulting cluster has the lowest radius.
(Or we could combine the clusters whose result has the lowest average distance between a point and the centroid.)
(Or we could use the sum of the squares of the distances between the points and the centroids.)

④ The **<u>diameter</u>** of a cluster is the maximum distance between any two points of the cluster.
We may choose to merge those clusters whose resulting cluster has the smallest diameter.

5. Alternative Rules for Controlling Hierarchcial Clustering

- **Another <u>Option 2</u> for Hierarchical Clustering**

For predetermined k (k for the number of clusters),

① Stop if the diameter of the cluster that results from the best merger exceeds a threshold.

② Stop if the density of the cluster that results from the best merger is below some threshold.

Density can be defined in many ways, but basically it should be the number of cluster points per unit volume of the cluster.

③ Stop when there is evidence that the next pair of clusters to be combined yields a bad cluster.

6. Hierarchical Clutering in Non - Euclidean Spaces

- **No Centroids in Non - Euclidean Spaces**

=> We need to use some distance measures that is computed from points (Jaccard, Cosine or Edit Distance ...).

We need to pick one of the points of the cluster itself to represent the cluster.

- **Clutroid**

A clustroid is a representative point within a cluster,

chosen to minimize certain distance criteria relative to other points in the cluster.

Common Criteria for Choosing Clustroids

① Minimize the sum of the distances to the other points in the cluster.

② Minimize the maximum distance to any other points in the cluster.

③ Minimize the sum of squares of distances to other points in the cluster.

▪ Example : Choosing the Clustroid

Cluster Points: `abcd`, `aecdb`, `abecb`, `ecdab`

- **Distance Matrix:**

	abcd	aecdb	abecb	ecdab
abcd	-	3	3	5
aecdb	3	-	2	2
abecb	3	2	-	4
ecdab	5	2	4	-

- **Criteria Results:**

Point	Sum of Distances	Max Distance	Sum of Squares
abcd	11	5	43
aecdb	7	3	17
abecb	9	4	29
ecdab	11	5	45

- **Result**: Based on all criteria, `aecdb` is chosen as the **clustroid**.

- **Measuring Distances Between Clusters**

After identifying clustroids, distances between clusters can be computed using following methods :

① Distance Between Clustroids

- Use the distance between the two clustroids as the cluster distance.

② Average Pairwise Distance

- Average the distances between all pairs of points from the two clusters.

③ Minimum Pairwise Distance

- Minimum distance between any pair of points, one from each cluster.

6. Hierarchical Clutering in Non - Euclidean Spaces

- **Cluster Density in Non - Euclidean Spaces**

① Radius

- Define radius using the clustroid, with measures consistent with the clustroid selection criterion.

② Diameter

- Maximum distance between any two points in the cluster.

- **Stopping Criteria**

① Threshold - Based Merging

- Stop merging when the smallest inter-cluster distance exceeds a predefined theshold.

② Number of Clusters

- Stop when a specific number of clusters is reached.

③ Cluster Adequacy

- Stop when clusters exceed a size, diameter or density constraint.

Thank You